

Reproduction of Broadcom/csg-htsim: NDP sequential all-to-all on a 1024-node fat tree

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Independent reproduction · Artifact evaluation · Networking

Abstract

Built htsim at the pinned commit and ran the shipped experiment experiments/fattree-all-to-all/sequential_512conns: a 512-host sequential (shift) all-to-all with NDP and per-packet spraying on a 1024-node three-tier fat tree, 1 MB flows, seed 13, using the repository's own run.sh and post-processing scripts. The result of interest is the self-synchronization described in the experiment's README: how many senders converge on a single receiver. The final gnuplot step was skipped (gnuplot not installed); the older example programs under sim/datacenter/old_tests do not compile at this commit and were not used. Single-threaded simulation on an Intel Xeon W-2145.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative. This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00027v1>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Sequential all-to-all, 512 hosts	2026-10-08	1	1	Signed by the K-Veritas server

2 Results

r1: Sequential all-to-all, 512 hosts

Run	Metric	Final value	Step
1	flows_finished	261632	
1	packets_new	6.5408e+07	
1	packets_retransmitted	1.74483e+06	

Run	Inline claim	Value	Phase
1	max_senders_per_receiver	57	
1	last_flow_finish_us	135067	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	8856.8 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	56631af52f0b1734aafd115cfd94655bd54b7f90114a0f805372cd94b23d358d	5.6 MB
r1.kvbundle.zip	44585e956d55b8e27b5baeae8fd83f2a07f33683c3c295fa17b2125dd2371c12	654 KB

5 Links and licence

Reproduction of: <https://github.com/Broadcom/csg-htsim/tree/841d9e7be46bb968eece766aa4b6c044c7799f67>

Record: CC-BY 4.0. Code: BSD-2-Clause, as declared by the authors. Tags: network-simulation, ndp, datacenter, congestion-control.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Sequential all-to-all, 512 hosts), kv:2610.00027v1/r1. Data hash 02accad8d67271c8c02c4dc3231ac875bc0f56a60f58cbd1b85ef7db7b1f5a9e.
<https://kveritas.org/records/2610.00027v1/r1.pdf>